

# Project 2: YSU Quantitative Finance

Aramayis, Armen, Tigran

Yerevan State University

June - 17, 2026

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# Bayesian Updates: Bernoulli Data

Model each binary outcome  $X_i \in \{0, 1\}$  (e.g. is the daily return positive?) as  $X_i \stackrel{iid}{\sim} \text{Bernoulli}(p)$ .

**Prior:**  $p \sim \text{Beta}(a, b)$ , default  $a = b = 1$  (Uniform)

**Posterior** (conjugacy), with  $s = \sum x_i$  successes out of  $n$ :

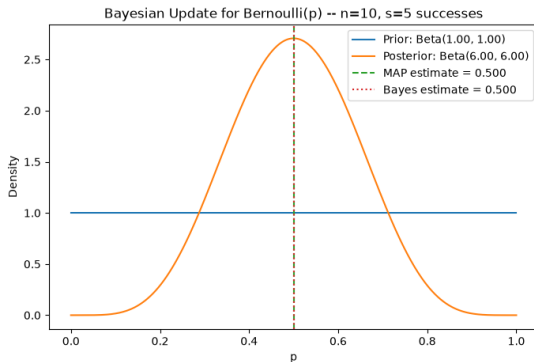
$$p \mid \text{data} \sim \text{Beta}(a + s, b + n - s)$$

**Estimates** from posterior  $\text{Beta}(a', b')$ :

$$\hat{p}_{\text{MAP}} = \frac{a' - 1}{a' + b' - 2}, \quad \hat{p}_{\text{Bayes}} = \frac{a'}{a' + b'}$$

# Bernoulli Example: Toy Data

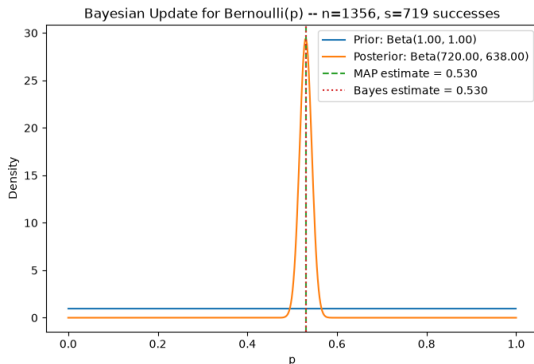
Prior Beta(1, 1), data 1100101100 ( $n = 10$ ,  $s = 5$  successes)



Posterior Beta(6, 6),  $\hat{p}_{\text{MAP}} = \hat{p}_{\text{Bayes}} = 0.500$

# Bernoulli Example: Is AAPL's Daily Return Positive?

$X_t = 1$  if AAPL's daily log return is positive, 0 otherwise. Prior Beta(1, 1),  $n = 1356$  days,  $s = 719$  positive days.



Posterior Beta(720, 638),  $\hat{p}_{\text{MAP}} = \hat{p}_{\text{Bayes}} \approx 0.530$

- AAPL closes up on  $\approx 53\%$  of days – slightly more often than not

# Bayesian Updates: Normal Data

Model  $X_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$  with **known**  $\sigma$  (e.g. the mean daily return  $\mu$  of a stock).

**Prior:**  $\mu \sim N(\mu_0, \sigma_0^2)$ , default  $\mu_0 = 0$ ,  $\sigma_0 = 1$

**Posterior** (conjugacy), with sample mean  $\bar{x}$ :

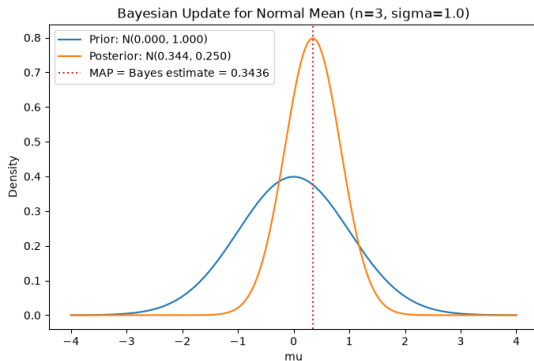
$$\mu \mid \text{data} \sim N(\mu_n, \sigma_n^2), \quad \frac{1}{\sigma_n^2} = \frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}, \quad \mu_n = \sigma_n^2 \left( \frac{\mu_0}{\sigma_0^2} + \frac{n\bar{x}}{\sigma^2} \right)$$

For a Normal posterior, mode = mean:

$$\hat{\mu}_{\text{MAP}} = \hat{\mu}_{\text{Bayes}} = \mu_n$$

# Normal Example: Toy Data

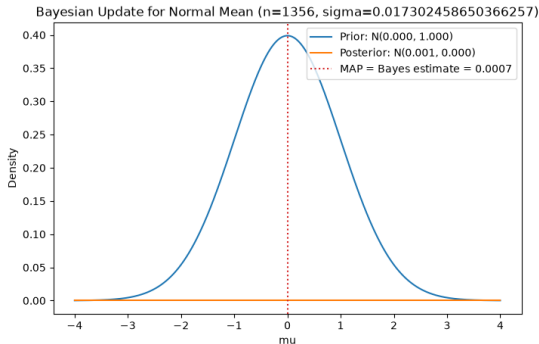
$\sigma = 1$ , prior  $N(0, 1)$ , data 0.123, 1.3812,  $-0.13$  ( $n = 3$ )



Posterior  $N(0.344, 0.25)$ ,  $\hat{\mu}_{\text{MAP}} = \hat{\mu}_{\text{Bayes}} = 0.344$

# Normal Example: Mean of AAPL Daily Returns

$\sigma$  fixed at the sample std of AAPL's log returns, prior  $N(0, 1)$ ,  $n = 1356$



Posterior mean  $\hat{\mu}_{\text{Bayes}} \approx 0.0007$  – with  $n = 1356$ , the data completely dominates the (wide) prior



## Supplementary: Does the Posterior Forget the Prior?

Known true  $\mu = 2$ ,  $\sigma = 1$ ; prior  $N(0, 1)$  (deliberately off-center). Generate  $n$  points from the true distribution and compute the posterior.

`../figures/normal_convergence.png`

- **Tickers:** ^IXIC (NASDAQ Composite), AAPL (Apple), GOOGL (Alphabet)
- **Period:** 2021-01-01 to 2026-06-01, daily data ( $\approx 1356$  observations)
- **Variable:** daily log returns

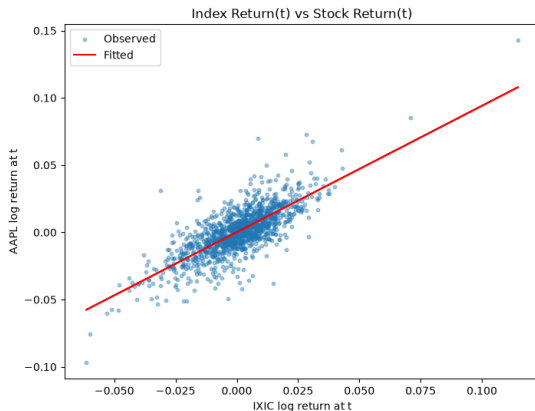
$$r_t = \log \left( \frac{P_t}{P_{t-1}} \right)$$

- **Method:** Ordinary Least Squares (OLS), via `statsmodels`

# Model 1: Index Return( $t$ ) $\rightarrow$ Stock Return( $t$ )

**Question:** Can market index returns explain individual stock returns?

$$\text{AAPL Return}_t = \beta_0 + \beta_1 \cdot (\text{NASDAQ Return})_t + \varepsilon_t$$



# Model 1: Results

$$\widehat{\text{AAPL}}_t = 0.00015 + 0.940 \cdot \text{IXIC}_t$$

| Coefficient                           | Estimate | <i>p</i> -value              |
|---------------------------------------|----------|------------------------------|
| $\beta_0$ (intercept)                 | 0.00015  | 0.625 (not significant)      |
| $\beta_1$ (IXIC <sub><i>t</i></sub> ) | 0.940    | $\approx 5 \times 10^{-260}$ |

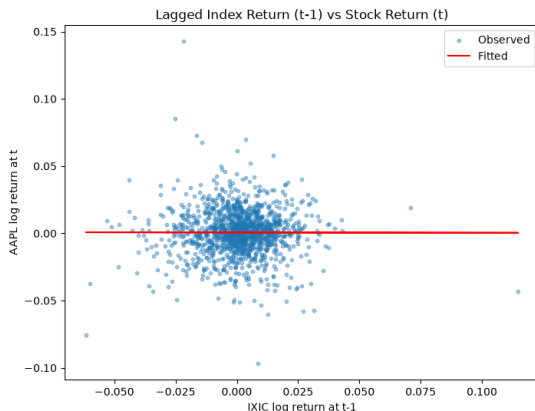
$R^2 = 0.584$ ,    Adjusted  $R^2 = 0.583$

- AAPL's daily return moves almost one-for-one with the NASDAQ ( $\beta_1 \approx 1$ )
- $\approx 58\%$  of AAPL's return variance is explained by the index return
- Residuals: fat-tailed (kurtosis  $\approx 6$ ), Durbin-Watson  $\approx 1.84$

## Model 2: Lagged Index Return( $t-1$ ) $\rightarrow$ Stock Return( $t$ )

**Question:** Can past returns help predict future returns?

$$\text{AAPL Return}_t = \beta_0 + \beta_1 \cdot (\text{NASDAQ Return})_{t-1} + \varepsilon_t$$



## Model 2: Results

$$\widehat{\text{AAPL}}_t = 0.00066 - 0.0024 \cdot \text{IXIC}_{t-1}$$

| Coefficient                             | Estimate | <i>p</i> -value         |
|-----------------------------------------|----------|-------------------------|
| $\beta_0$ (intercept)                   | 0.00066  | 0.159 (not significant) |
| $\beta_1$ (IXIC <sub><i>t</i>-1</sub> ) | -0.0024  | 0.943 (not significant) |

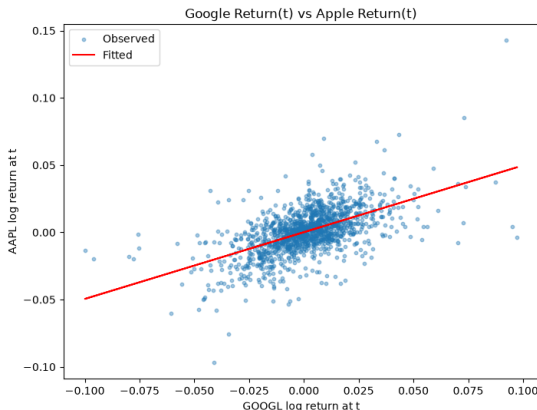
$R^2 \approx 0.0000$ ,    Adjusted  $R^2 \approx -0.0007$

- Yesterday's NASDAQ return has **no** predictive power for today's AAPL return
- Consistent with the (weak-form) Efficient Market Hypothesis

## Model 3: Google Return( $t$ ) $\rightarrow$ Apple Return( $t$ )

**Question:** Can one stock's returns explain another stock's returns?

$$\text{AAPL Return}_t = \beta_0 + \beta_1 \cdot (\text{GOOGL Return})_t + \varepsilon_t$$



## Model 3: Results

$$\widehat{\text{AAPL}}_t = 0.00013 + 0.496 \cdot \text{GOOGL}_t$$

| Coefficient                            | Estimate | <i>p</i> -value                |
|----------------------------------------|----------|--------------------------------|
| $\beta_0$ (intercept)                  | 0.00013  | 0.750 (not significant)        |
| $\beta_1$ (GOOGL <sub><i>t</i></sub> ) | 0.496    | $\approx 6.5 \times 10^{-113}$ |

$$R^2 = 0.314, \quad \text{Adjusted } R^2 = 0.313$$

- A 1% same-day move in GOOGL is associated with a  $\approx 0.5\%$  move in AAPL
- $\approx 31\%$  of AAPL's variance explained – less than the index (Model 1, 58%)
- Suggests a common market/sector factor, not a direct causal link



# Stage 1: Summary

| Model | Predictor    | $R^2$ | Significant? |
|-------|--------------|-------|--------------|
| 1     | $IXIC_t$     | 0.584 | Yes          |
| 2     | $IXIC_{t-1}$ | 0.000 | No           |
| 3     | $GOOGL_t$    | 0.314 | Yes          |

- Same-day market index return strongly explains AAPL's return ( $R^2 = 0.58$ )
- Lagged (past) returns carry **no** predictive information – market efficiency
- Same-day peer-stock (GOOGL) return also explains a sizeable share ( $R^2 = 0.31$ ), likely via shared exposure to the market factor

# Model A: Full Lagged Model

**Question:** Can past returns predict future returns?

$$\begin{aligned} \text{AAPL}_t = & \beta_0 + \beta_1 \text{IXIC}_{t-1} + \beta_2 \text{AAPL}_{t-1} + \\ & + \beta_3 \text{MA}_{20} + \beta_4 \text{MA}_{50} + \beta_5 \text{MA}_{200} + \\ & + \beta_6 \text{VOL}_{20} + \beta_7 \text{VOL}_{20} + \beta_8 \text{VOL}_{20} + \\ & + \beta_9 \Delta \text{Volume}_{t-1} + \beta_{10} \text{VolRatio}_{t-1} + \varepsilon_t \end{aligned}$$

$$R^2 = 0.010, \quad \text{Adjusted } R^2 = 0.001, \quad F\text{-test } p = 0.336$$

- Overall model is **not statistically significant**
- Only VolumeChg\_lag1 is individually significant ( $p = 0.031$ )

## Model B: Reduced Model

$$AAPL_t = \beta_0 + \beta_1 \cdot IXIC_{t-1} + \beta_2 \cdot AAPL_{t-1} + \varepsilon_t$$

| Coefficient           | Estimate | <i>p</i> -value |
|-----------------------|----------|-----------------|
| $\beta_0$ (intercept) | 0.0007   | 0.201           |
| $\beta_1$ (IXIC_lag1) | -0.0007  | 0.383           |
| $\beta_2$ (AAPL_lag1) | 0.0009   | 0.243           |

$$R^2 = 0.001, \quad \text{Adjusted } R^2 = -0.001, \quad F\text{-test } p = 0.506$$

- Mirrors Stage 1 Model 2: lagged index return still has zero predictive power
- Own lagged return adds nothing either

# Model C: Volume-Focused Model

**Question:** Can trading volume explain/predict returns?

$$AAPL_t = \beta_0 + \beta_1 \cdot AAPL_{t-1} + \beta_2 \cdot \Delta Volume_{t-1} + \beta_3 \cdot VolRatio_{t-1} + \varepsilon_t$$

| Coefficient                  | Estimate | <i>p</i> -value |
|------------------------------|----------|-----------------|
| $\beta_0$ (intercept)        | 0.0007   | 0.200           |
| $\beta_1$ (AAPL_lag1)        | 0.0004   | 0.471           |
| $\beta_2$ (VolumeChg_lag1)   | -0.0013  | 0.032           |
| $\beta_3$ (VolumeRatio_lag1) | 0.0009   | 0.158           |

$R^2 = 0.005$ ,    Adjusted  $R^2 = 0.002$ ,     $F$ -test  $p = 0.152$

- Overall model not significant, but VolumeChg\_lag1 is ( $p = 0.032$ )
- With 10 predictors tested in Model A, one hit at  $p \approx 0.03$  is roughly what chance alone would produce

# Model Comparison

| Model              | $R^2$ | Adj. $R^2$ | AIC     | BIC     |
|--------------------|-------|------------|---------|---------|
| A (full lagged)    | 0.010 | 0.001      | −6053.4 | −5997.9 |
| B (reduced)        | 0.001 | −0.001     | −6059.5 | −6044.3 |
| C (volume-focused) | 0.005 | 0.002      | −6061.4 | −6041.2 |

- Model C has the lowest AIC; Model B has the lowest BIC – the differences are negligible
- Model A is worst on both criteria (most parameters, no gain in fit)
- **None** of the three models pass the overall  $F$ -test at the 5% level

# Residual Diagnostics (Model C)

- Durbin–Watson  $\approx 2.00$  – no autocorrelation in residuals
- Jarque–Bera strongly rejects normality; kurtosis  $\approx 8.7$
- Same fat-tailed pattern as Stage 1 – typical of daily financial returns, and largely independent of which (weak) model is used

## Stage 2/3: Summary

- **Market index returns:** explain AAPL *contemporaneously* ( $R^2 = 0.58$ , Stage 1), but the lagged index has **zero** predictive power ( $R^2 \approx 0$ )
- **Past returns:** across Models A, B, C,  $R^2 \leq 1\%$  and the  $F$ -test is never significant – past returns do **not** predict future returns
- **Trading volume:** VolumeChg\_lag1 is marginally significant alone ( $p \approx 0.03$ ), but the overall volume model is not ( $p = 0.152$ ) – weak, inconclusive evidence
- **Conclusion:** results are consistent with (weak-form) market efficiency – information is priced in immediately, leaving no exploitable pattern in lagged data

# Defining Market Regimes

Classify each day's NASDAQ Composite (^IXIC) into **Bull**, **Neutral**, or **Bear** using the 20-day rolling cumulative log return

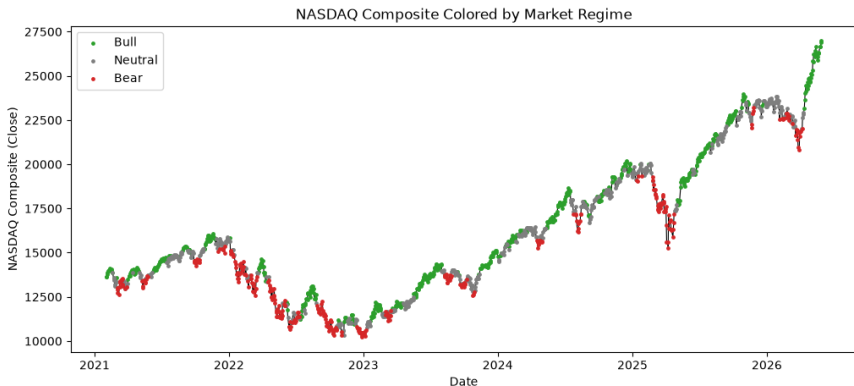
$$R_t = \sum_{i=0}^{19} r_{t-i}, \quad \theta = 0.5 \cdot \text{std}(R_t)$$

$$\text{Regime}_t = \begin{cases} \text{Bull} & R_t > \theta \\ \text{Bear} & R_t < -\theta \\ \text{Neutral} & \text{otherwise} \end{cases}$$

| Regime   | Bull        | Neutral     | Bear        |
|----------|-------------|-------------|-------------|
| Days (%) | 554 (41.4%) | 465 (34.8%) | 318 (23.8%) |



# NASDAQ Composite Colored by Regime



# Transition Probability Matrix

$$P_{ij} = P(\text{Regime}_{t+1} = j \mid \text{Regime}_t = i)$$

$$P = \begin{pmatrix} 0.879 & 0.121 & 0.000 \\ 0.144 & 0.751 & 0.105 \\ 0.000 & 0.154 & 0.846 \end{pmatrix} \quad (\text{rows/cols: Bull, Neutral, Bear})$$

- All three regimes are highly **persistent** (diagonal  $\approx 0.75$ – $0.88$ )
- $P(\text{Bull} \rightarrow \text{Bear}) = P(\text{Bear} \rightarrow \text{Bull}) = 0$  – the market always passes through Neutral

# Stationary Distribution

$$\pi P = \pi, \sum_i \pi_i = 1$$

| Regime  | Stationary $\pi$ | Observed frequency |
|---------|------------------|--------------------|
| Bull    | 0.414            | 0.414              |
| Neutral | 0.348            | 0.348              |
| Bear    | 0.238            | 0.238              |

- Stationary distribution matches observed frequencies almost exactly – the chain is essentially already at its long-run distribution

# Predicting the Next Regime

Given the current regime  $i$ , the most probable next regime is  $\arg \max_j P_{ij}$ .

- Last observed regime: **Bull**
- $P(\text{Bull} \rightarrow \text{Bull}) = 0.879$ ,  $P(\text{Bull} \rightarrow \text{Neutral}) = 0.121$ ,  
 $P(\text{Bull} \rightarrow \text{Bear}) = 0.000$
- Predicted next regime: **Bull**

# Simulating the Markov Chain

Simulate a regime sequence of the same length from  $P$  and compare to the observed data.

| Regime  | Frequency |           |
|---------|-----------|-----------|
|         | Observed  | Simulated |
| Bull    | 0.414     | 0.476     |
| Neutral | 0.348     | 0.301     |
| Bear    | 0.238     | 0.223     |

| Regime  | Avg. run length (days) |           |
|---------|------------------------|-----------|
|         | Observed               | Simulated |
| Bull    | 8.1                    | 8.8       |
| Neutral | 4.0                    | 3.3       |
| Bear    | 6.5                    | 6.0       |

- A first-order Markov chain reproduces both the regime frequencies and the average regime durations reasonably well

## Part III: Summary

- Market regimes (Bull/Neutral/Bear) are **highly persistent**, and the chain is essentially at its stationary distribution
- The model predicts the current Bull regime will most likely continue
- However, persistence is largely **mechanical**:  $R_t$  is a 20-day rolling sum, so  $R_t$  and  $R_{t-1}$  share 19 of 20 terms and are highly correlated by construction
- Consistent with Stage 1: *daily returns* themselves remain essentially unpredictable (weak-form efficiency); only the smoothed regime label is persistent